



Sri Lanka Institute of Information Technology

B.Sc. Honours Degree in Information Technology

Specialized in Information Technology/ Data Science

Final Examination  
Year 3, Semester 1 (2024)

IT3030 – Programming Applications and  
Frameworks

Duration: 2 Hours

May/ June 2024

Instructions to Candidates:

- ◆ This paper has four questions.
- ◆ This paper has a ten-minute reading time allocation.
- ◆ Answer all questions in the booklet given.
- ◆ The total marks for the paper is 100.
- ◆ This paper contains 4 pages, including the cover page.
- ◆ Electronic devices capable of storing and retrieving text, including calculators and mobile phones are not allowed.

**Question 1****(25 marks)**

1. Briefly discuss how software frameworks help in engineering good software.  
(2 marks)
2. Identify a situation in which you used a software framework (name the framework) and discuss the reasons for utilizing that framework in that situation rather than building a solution from scratch.  
(5 marks)
3. A common habit among some software developers nowadays is to learn a software framework and then try to apply the same software framework to solve any issue that they may encounter. Do you think this practice is good? Briefly discuss and justify your position with three (3) reasons.  
(10 marks)
4. With a suitable example discuss how the global control flow of an application built using a software framework differs from an application written from scratch, identifying the potential benefits a developer using a software framework may have over one who does not.  
(8 marks)

**Question 2****(25 marks)**

1. Briefly discuss the importance of Version control systems (VCS) in modern software development in your own words.  
(2 marks)
2. With a suitable example, differentiate between Localized version control systems (LVCS) and Distributed version control systems (DVCS) with at least four (4) points. Clearly highlight the differences in approaches to version controlling in each method.  
(4 marks)
3. "Maintaining a central remote repository even though it is not mandatory is a good practice when version controlling a software project with Git, especially when multiple developers are involved." Do you agree with the above statement? Give reasons.  
(6 marks)
4. Company Y is a startup venture launched by a group of six programmers fresh out of university. Their first project involves building a mobile application for managing personal finances. Despite their lack of experience with VCS, they opt to employ Git. They have planned to maintain their individual code in long-living individual branches under their names in local as well as remote repositories until it is time to integrate, at which point all branches would be reviewed and merged with the main development branch. This is to avoid propagating any issues in individual developments to the main branch. You, an IT expert, have been asked to evaluate their strategy.

- a. With reasons, discuss why it is better to frequently merge code from individual branches to the main development branch rather than follow the approach that they have planned to use.

(4 marks)

- b. A git concept called “Pull Requests (PR)” can help them to overcome the problem they are concerned about by allowing them to review changes while merging individual changes to the main development branch continuously without waiting until the time to integrate. However, the developers are quite skeptical of this. Discuss with an example based on a real-world software development environment, how Pull Requests (PR) can help minimize problems.

(4 marks)

- c. To keep the flow of daily development operations running smooth, the developers have thought to adopt a Git workflow. They also have plans to grow the organization considerably in the next two years. GitFlow and GitHub flow have caught their attention as being suitable for their operation. Since GitFlow seems very comprehensive, they are thinking of using it for all projects including future projects.

Considering the plans of the company, discuss and justify with reasons if using GitFlow is a good idea or not in your view.

(5 marks)

### Question 3

(25 marks)

1. Briefly discuss the importance of Application Programming Interfaces (API) in enabling collaboration among different applications.

(3 marks)

2. Representational State Transfer (REST) is a very popular style of APIs nowadays. Briefly explain what REST is, and clearly highlight what makes an API, a REST API.

(4 marks)

3. “Client-server architecture” is an important constraint in REST. Briefly explain this constraint, and then with an example clearly show how this constraint helps with scaling the backend (server-side) of a web application.

(5 marks)

4. Hypermedia as the engine of application state (HATEOAS) is an important constraint in REST.

- a. Briefly explain this concept in your own words.

(2 marks)

- b. Briefly discuss the benefits a web client may gain when consuming an API aligning with the above constraint.

(3 marks)

- c. With an example, clearly show the difference between a **response** from a fully REST compliant API vs. an API which is not REST compliant. You can assume that it is the same endpoint in the same system responding in both instances – in one instance, the system has a REST compliant API and in the other instance,

it does not. You can also assume that both systems respond with JSON (JavaScript Object Notation). Please provide a small description explaining your example.

(4 marks)

5. It is a best practice to use correct HTTP methods, headers, and status codes according to the operation being performed on an API. Why do you think this is so? Justify your answer.

(4 marks)

#### **Question 4**

**(25 marks)**

1. Briefly explain the core DevOps principles.

(4 marks)

2. With appropriate justifications, discuss why DevOps requires a complete change of an IT organization's culture rather than just being a mere software development methodology.

(4 marks)

3. The upper management in a large-scale product-based software company is considering introducing DevOps to the company in hopes of addressing the current issues it faces with delivery and deployment of software using the traditional waterfall model. They encounter frequent delays in the Testing phase of the waterfall model as unexpected bugs are discovered.

- a. What would be some common issues that they could be facing? Identify and briefly discuss these issues.

(3 marks)

- b. "To successfully practice DevOps, the organization needs to break down the silos and get everyone on the same team". Critically discuss your views on this statement.

(4 marks)

- c. Automation is a key principle in DevOps. Briefly discuss how various forms of Automation can be applied in minimizing the problems they encounter at present.

(6 marks)

- d. Some members of the organization also think that they need to overhaul their Software Quality Assurance process given the issues that they face rather than just depend on Testing. Do you agree with this? Give reasons.

(4 marks)